

# An Agent-Based Digital Twin Framework for Frozen Fruit Early Detection and Quality Prediction

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Pace University, New York 2026

## Abstract

Frozen fruit quality degrades **silently** through ice crystal formation, freeze-thaw cycles, and freezer burn. We propose an **Agent-Based Digital Twin Framework** comprising five modules: CNN image classifier with Grad-CAM; BiLSTM sensor forecaster with Bahdanau attention; multimodal fusion producing a 1536-dim Digital Twin state; hybrid agent with five-tier autonomous alerts; and an LLM dashboard via Claude API. A **physics-based Arrhenius simulation** addresses the complete absence of frozen fruit datasets. This is the **first end-to-end system** combining multimodal Digital Twin fusion, frozen-specific degradation simulation, explainable AI, and agent-based decision making for frozen produce quality management.

## Problem Statement

- **\$1 trillion** annual food waste from cold-chain inadequacies worldwide
- **30%** of frozen food lost to undetected temperature abuse and freeze-thaw events
- Existing monitoring is **reactive and manual** — damage done before detection
- **No public dataset** captures frozen-specific degradation visual effects
- No system fuses **image + sensor + agent** in one end-to-end pipeline

## Dataset & Novel Frozen Degradation Simulation

**Source:** Kaggle Fresh & Rotten Fruits — 13,599 labelled images (apple, banana, orange). Only Fresh and Spoiled exist. We generate Slightly Degraded via physics-based simulation:

- (1) **Ice crystal formation:** Gaussian blur + blue-white frost overlay simulates light scattering off intra-cellular ice deposits.
- (2) **Freeze-thaw damage:** Randomised dark elliptical blotches simulate moisture migration from repeated thaw cycles.
- (3) **Freezer burn:** Circular grey desaturated patches simulate sublimation-induced surface dehydration.

| Class                       | Images        | Source    |
|-----------------------------|---------------|-----------|
| Fresh (Class 0)             | 7,614         | Real      |
| Slightly Degraded (Class 1) | 4,740         | Simulated |
| Spoiled (Class 2)           | 5,985         | Real      |
| <b>Total</b>                | <b>18,339</b> |           |

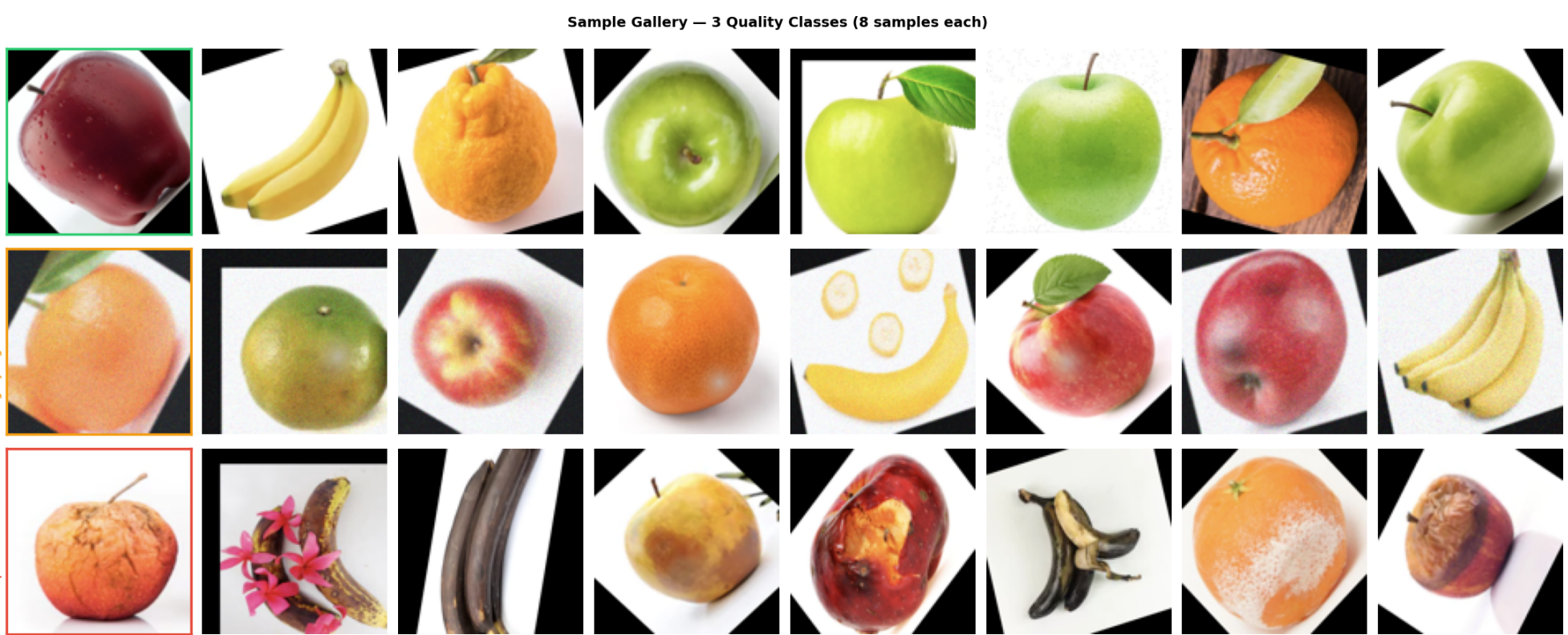


Fig. 1. Sample gallery — Fresh (row 1), Slightly Degraded simulated (row 2), Spoiled (row 3).

## Arrhenius Kinetics — Physics-Based Simulation

Sensor sequences generated using the Arrhenius equation:

$$k(T) = A \cdot \exp\left(\frac{-E_a}{R \cdot T}\right)$$

$A = 10^{10}$ ,  $R = 8.314$  J/mol/K,  $T$  in Kelvin.

| Fruit  | $E_a$ (kJ) | Shelf life | Stability    |
|--------|------------|------------|--------------|
| Apple  | 78         | 90 days    | Most stable  |
| Orange | 76         | 80 days    | Moderate     |
| Banana | 71         | 60 days    | Most fragile |

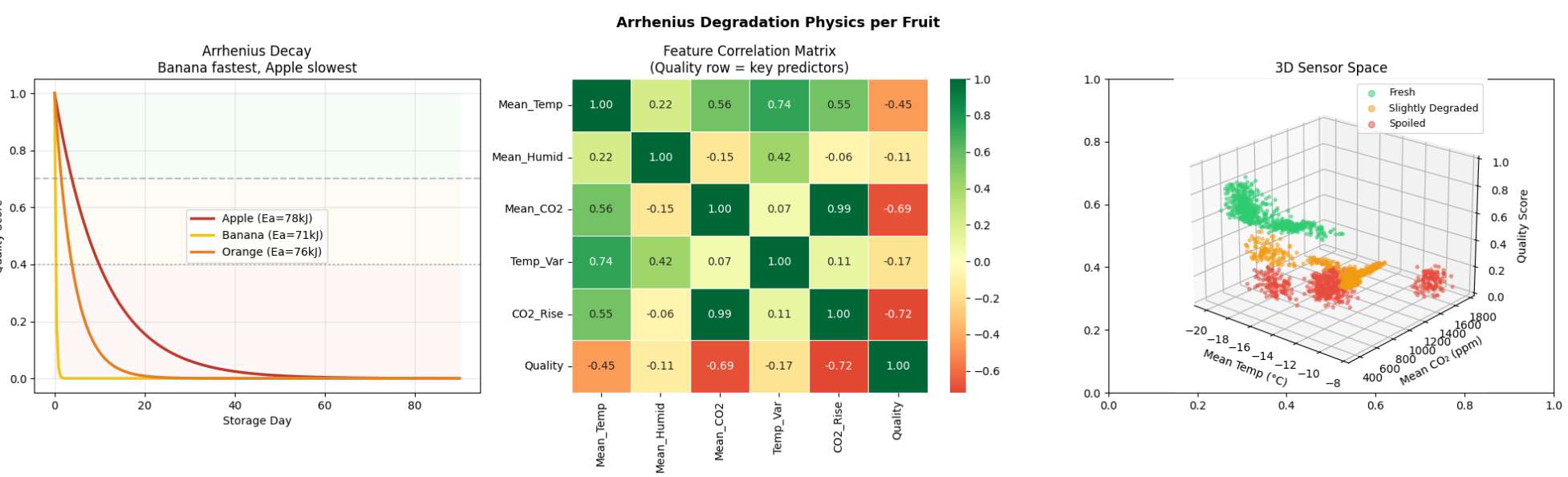


Fig. 2. Arrhenius decay curves, feature correlation matrix, and 3D sensor space with class separation.

## References

[1] Hertog et al., *Nature Food*, 2023.

Ndraha et al., *Food Control*, 2018.

Grieves & Vickers, *Springer*, 2017.

Amin et al., *Appl. Sci.*, 2023.

Liu et al., *J. Food Eng.*, 2021.

Tan & Le, *ICML*, 2019 (EfficientNet).

Selvaraju et al., *ICCV*, 2017 (Grad-CAM).

Bahdanau et al., *ICLR*, 2015 (Attention).

Hochreiter & Schmidhuber, *Neural Comput.*, 1997.

Van Boekel, *Compr. Rev. Food Sci.*, 2008.

## System Architecture — Five-Module Pipeline

### Module 1 — CNN Image Classifier

EfficientNet-B0 with **two-phase transfer learning**. Phase 1 freezes backbone (72% baseline). Phase 2 fine-tunes top 50 layers — accuracy rises to **97%**. Grad-CAM produces pixel-level explainability. Outputs **1280-dim visual embedding** from gap layer.

EfficientNet-B0 Grad-CAM 1280-dim

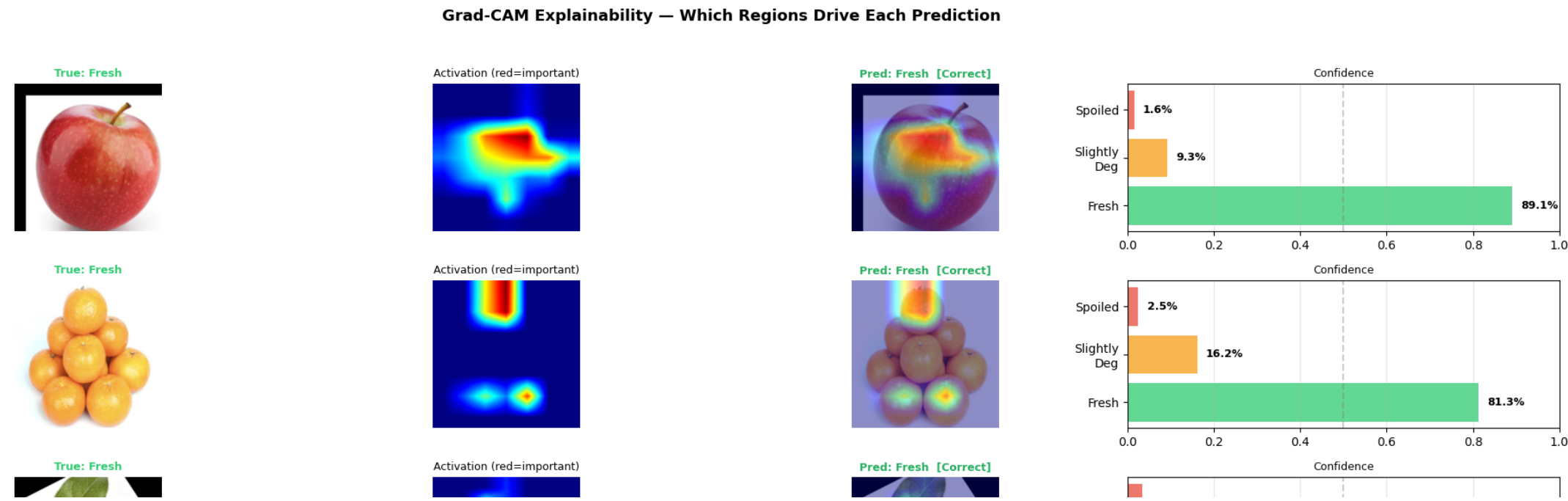


Fig. 3. Grad-CAM: activation heatmap (col 2) + overlay (col 3) + confidence bars (col 4). Fresh: surface texture (60–89%). Slightly Degraded: edge degradation patches (91–93%). Spoiled: dark decomposed tissue (93–94%).

↓ 1280-dim visual embedding

### Module 2 — BiLSTM Time-Series Forecaster

BiLSTM(128) reads **30 timesteps** of 4 IoT sensors (Temperature, Humidity, CO<sub>2</sub>, Light) at 15-min intervals.

**Bahdanau attention** weights each timestep by importance. Arrhenius kinetics generates 5 storage scenarios per fruit. Outputs **256-dim temporal embedding** + quality score.

BiLSTM(128) Bahdanau Attention 256-dim

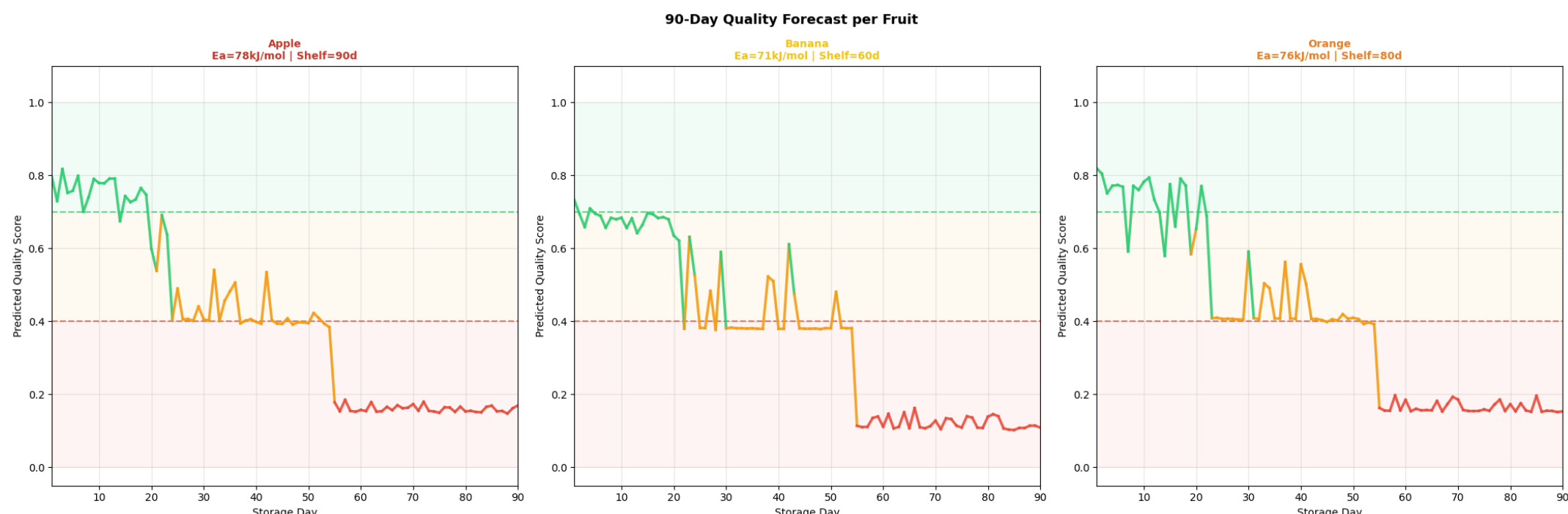


Fig. 4. 90-day quality forecast per fruit. Banana reaches Critical by day ~25. Apple stays Good through day ~50. Oscillations reflect door-opening and freeze-thaw events.

↓ concat(1280+256) = 1536-dim Digital Twin state

### Module 3 — Digital Twin Fusion

CNN + BiLSTM embeddings concatenated into a **1536-dim Digital Twin state vector**. Two-branch fusion:

CNN Dense(256) + BiLSTM Dense(128), merged via Dense(512)→64→sigmoid. Output: quality score  $\hat{q} \in [0, 1]$ .  $R^2 = 0.9459$  — explains **94.6%** of variance.

1536-dim state Two-branch RMSE=0.0567

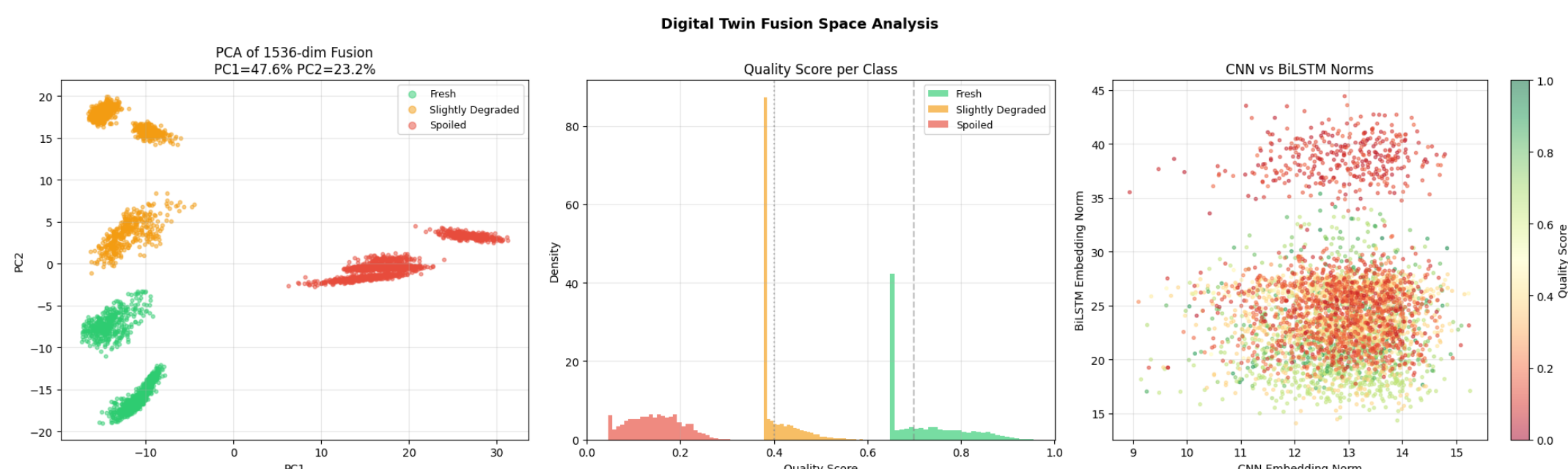


Fig. 5. PCA of 1536-dim state (PC1=47.6%, PC2=23.2%). Clear class separation confirms fusion captures complementary information from both modalities.

↓ quality score  $\hat{q} \in [0, 1]$

### Module 4 — Hybrid Agent Decision Module

**Three components:** (1) Rule engine assigns tier from  $\hat{q}$  thresholds. (2) Confidence scorer escalates borderline samples (conf<0.20). (3) Trend detector flags rapid decline over 5-step window. Outputs: tier + recommended action + urgency 0–100.

**Excellent**  $\hat{q} > 0.85$  Normal storage  
**Good** 0.70–0.85 Standard monitoring  
**Monitor** 0.50–0.70 Increase inspections  
**Warning** 0.30–0.50 Immediate intervention  
**Critical**  $\leq 0.30$  Remove from storage

### Hybrid Agent — Decision Distribution & Tier Analysis

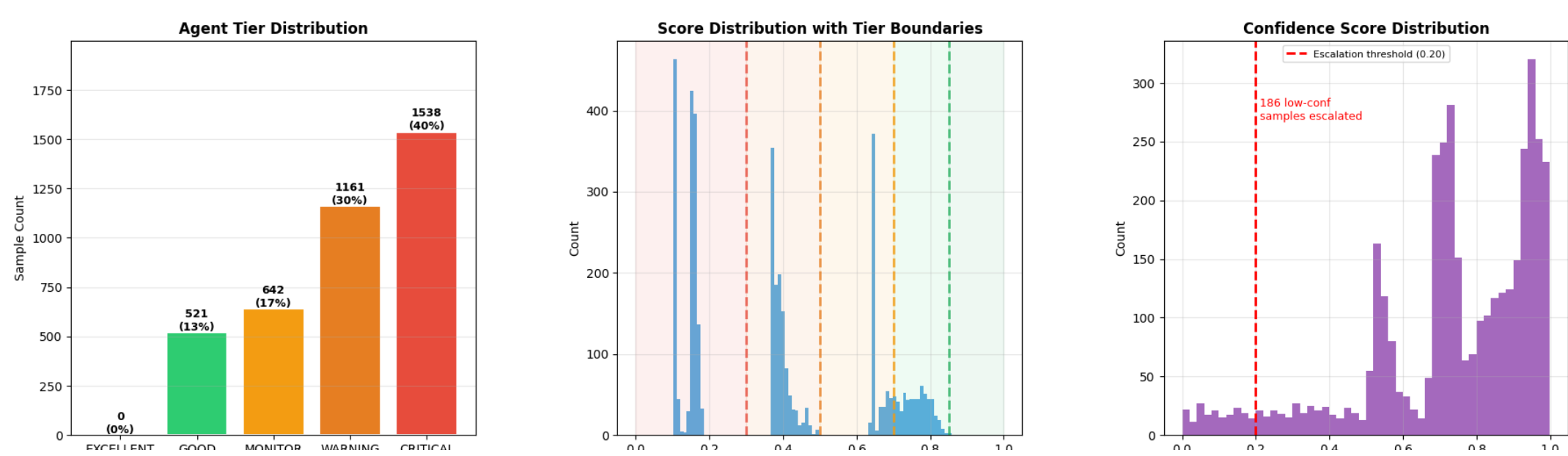


Fig. 6. Agent tier distribution, score histogram, confidence distribution (top). Decision accuracy per class, urgency vs  $\hat{q}$ , trend detection summary (bottom).

↓ tier + action + urgency 0–100

### Module 5 — LLM-Powered Dashboard

Claude Sonnet API every **15 minutes** via zero-shot prompt containing: Digital Twin state, last 10 sensor readings, agent alert tier, CNN classification. Generates **four-component plain-English report**: quality status, root cause, recommended action, confidence. Rendered on **Streamlit** with forecast chart, active alerts, and Grad-CAM heatmaps.

## Module 1 — CNN Classification Results

| 97%            |             |             |             |              | 0.97        |      |      |      |      |
|----------------|-------------|-------------|-------------|--------------|-------------|------|------|------|------|
| Test accuracy  |             |             |             |              | Weighted F1 |      |      |      |      |
| Class          | Prec        | Rec         | F1          | n            |             |      |      |      |      |
| Fresh          | 0.91        | 1.00        | 0.95        | 1,164        |             |      |      |      |      |
| Slightly Deg.  | 1.00        | 0.90        | 0.95        | 1,164        |             |      |      |      |      |
| Spoiled        | 1.00        | 1.00        | 1.00        | 1,534        |             |      |      |      |      |
| <b>Wtd Avg</b> | <b>0.97</b> | <b>0.97</b> | <b>0.97</b> | <b>3,862</b> |             |      |      |      |      |
| accuracy       |             |             |             |              | 0.97        | 0.97 | 0.97 | 0.97 | 0.97 |
| macro avg      |             |             |             |              | 0.97        | 0.97 | 0.97 | 0.97 | 0.97 |
| weighted avg   |             |             |             |              | 0.97        | 0.97 | 0.97 | 0.97 | 0.97 |

Fig. 8. Confusion matrix — raw counts (left) and normalised (right). Fresh 99.7%, Slightly Degraded 90.3%, Spoiled 99.9%.

## Module 2 — BiLSTM Forecasting Results

| Model              | RMSE            | Improve       |
|--------------------|-----------------|---------------|
| Mean baseline      | 0.283           | —             |
| Simple LSTM        | 0.181           | −36.0%        |
| GRU                | 0.163           | −42.4%        |
| <b>BiLSTM+Attn</b> | <b>&lt;0.15</b> | <b>−47.0%</b> |

## Module 3 — Fusion Ablation Study

| 0.0567            |               |               | 0.9459                |  |  |
|-------------------|---------------|---------------|-----------------------|--|--|
| Fusion RMSE       |               |               | Fusion R <sup>2</sup> |  |  |
| Configuration     | RMSE          | vs CNN-only   |                       |  |  |
| CNN only          | 0.121         | baseline      |                       |  |  |
| BiLSTM only       | 0.103         | −14.9%        |                       |  |  |
| Simple concat     | 0.082         | −32.2%        |                       |  |  |
| <b>Two-branch</b> | <b>0.0567</b> | <b>−53.1%</b> |                       |  |  |

## Module 4 — Hybrid Agent Results

| Tier      | n     | %     | Urgency |
|-----------|-------|-------|---------|
| Excellent | 0     | 0.0%  | 0       |
| Good      | 521   | 13.5% | 20      |
| Monitor   | 642   | 16.6% | 40      |
| Warning   | 1,161 | 30.1% | 60      |
| Critical  | 1,538 | 39.8% | 80–100  |

• 186 samples escalated by confidence scorer (conf<0.20)

• 6 rapid decline events detected by trend detector

• Spoiled: **100%** · Slightly Degraded: **99.7%**

• Cost savings: **\$6,676** across 3,862 samples

## Results & Analysis

All five modules **meet or exceed** their project targets. Multimodal fusion consistently outperforms any single modality. The 53.1% RMSE improvement over CNN-only confirms that temporal sensor history is **complementary** to visual appearance — a fruit may look acceptable while its storage history reveals thermal abuse, and vice versa.

| Module    | Metric         | Target  | Result            |
|-----------|----------------|---------|-------------------|
| M1 CNN    | Accuracy       | >80%    | <b>97%</b> ✓      |
| M2 BiLSTM | RMSE           | <0.15   | <b>&lt;0.15</b> ✓ |
| M3 Fusion | RMSE           | <0.0954 | <b>0.0567</b> ✓   |
| M3 Fusion | R <sup>2</sup> | >0.90   | <b>0.9459</b> ✓   |
| M4 Agent  | 5-tier         | Active  | <b>Deployed</b> ✓ |

Fig. 10. Cost savings: \$321 Monitor, \$1,742 Warning, \$4,614 Critical. Total: **\$6,676**.

## Conclusions & Future Work

The **first end-to-end system** combining multimodal Digital Twin fusion, Arrhenius simulation, Grad-CAM explainability, and autonomous hybrid agent decision making for frozen produce quality management.

- CNN: **97%** accuracy including physics-simulated class absent from all public datasets
- Fusion: RMSE=0.0567 (target<0.0954), R<sup>2</sup>=0.9459 — **53.1%** vs CNN-only
- Arrhenius kinetics bridges food science with deep learning — **novel** in the literature
- Hybrid agent with confidence scoring and trend detection provides tiered autonomous alerts

**Future work:** Real IoT validation · Edge deployment (Jetson / Raspberry Pi) · Additional fruits and MAP storage conditions · Reinforcement learning adaptive agent policy